

SASHWAT JAIN

AI/ML Engineer | GenAI | RAG | LLM Applications | MLOps

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Professional Summary

Results-driven AI/ML Engineer with 3+ years of experience designing, building, and deploying production-grade LLM applications, RAG pipelines, and document intelligence systems at enterprise scale. Proven track record of scaling AI services to 150K+ documents/month, migrating critical OCR workloads to cloud-native infrastructure, and implementing MLOps best practices with MLflow and CI/CD. Deep expertise in prompt engineering, vector search, embedding models, and LLM integration using OpenAI, LLaMA, and Mistral. Skilled at building AI-powered microservices with FastAPI and Docker across multi-client environments.

- **3+ years** experience in AI/ML engineering with deep focus on LLM applications, RAG, and Generative AI systems.
- Expertise in building and scaling AI microservices using FastAPI and Docker, serving 150K+ documents/month.
- Proven success in cloud migration, MLOps implementation (MLflow), and LLM guardrails for enterprise clients.
- Strong foundation in prompt engineering, semantic search, vector databases, and document intelligence pipelines.

Technical Skills

- **Programming Languages:** Python (Advanced), Java, Go
- **AI/ML & Generative AI:** Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Prompt Engineering, LLM Fine-tuning, Embeddings, Vector Databases, Semantic Search, Named Entity Recognition (NER), Document Intelligence, OCR
- **LLM Models & Frameworks:** OpenAI GPT-4/3.5, LLaMA 2/3, Mistral AI, LangChain, LlamaIndex, Hugging Face Transformers, guardrails.ai
- **MLOps & Cloud:** MLflow, Docker, CI/CD (GitHub Actions), Model Deployment, Model Monitoring, Google Cloud Platform (GCP), Outscale, AWS
- **Backend & APIs:** FastAPI, REST APIs, Microservices Architecture
- **Libraries & Tools:** PyTorch, scikit-learn, Tesseract OCR, Pandas, NumPy, YOLOv8

Professional Experience

Dassault Systèmes

Specialist Software Engineer — AI/ML

Pune, India
Mar 2025 – Present

- Architected LLM-based document intelligence workflows using Mistral AI and LLaMA 3 with MCP (Model Context Protocol) servers for structured tool calling, reducing manual extraction effort by approx. 60%.
- Designed an AI guardrails framework using `guardrails.ai` — including prompt injection detection, output filtering, and PII redaction — ensuring responsible and compliant AI usage across 10+ internal enterprise clients.
- Built scalable Retrieval-Augmented Generation (RAG) pipelines with vector embeddings and semantic search, improving domain-specific document query accuracy by 40% and significantly reducing LLM hallucination.
- Implemented end-to-end MLflow experiment tracking covering prompt versioning, model artifacts, and evaluation metrics across 20+ experiments, cutting iteration cycle time by approx. 35%.
- Led cloud migration of on-prem OCR services to Outscale, containerizing workloads with Docker and automating deployments via CI/CD pipelines — improving deployment frequency by 3x.
- Designed and maintained high-throughput FastAPI microservices processing **150K+ documents/month** across enterprise clients, achieving 99.9% uptime SLA.
- Collaborated cross-functionally with product, DevOps, and data teams to integrate AI/ML capabilities into existing enterprise workflows, reducing time-to-production for new AI features by approx. 40%.

Dassault Systèmes

Associate Software Engineer — AI/ML

Pune, India
Jul 2023 – Mar 2025

- Built a production-ready Image-to-Text OCR service using Tesseract and Python, exposed via FastAPI REST APIs and consumed by 5+ downstream services.
- Developed a Digital Wallet Proof-of-Concept leveraging LLaMA-2 to extract 5+ structured entities (merchant, amount, date, category, currency) from receipts with over 85% extraction accuracy.
- Designed modular OCR pipelines with configurable preprocessing (deskewing, noise removal), parsing, and structured extraction stages — improving OCR accuracy by 25%.
- Delivered production-grade REST APIs with comprehensive unit/integration testing, error handling, and API documentation, reducing post-release bug rate by 30%.

Key Projects

RAG-Based Chatbot with Document Ingestion Pipeline

- Designed a full RAG pipeline: document ingestion, chunking, embedding generation, vector store indexing, and LLM response generation — reducing hallucination rate by over 45% vs. baseline LLM on domain-specific queries.
- Integrated OpenAI embeddings with a vector database for semantic search; built a modular, testable architecture supporting plug-and-play LLM backends (GPT-4, LLaMA, Mistral).

LLM-Based Multilingual Translation Service

- Developed a high-accuracy translation microservice using OpenAI APIs with advanced prompt engineering for context-aware, idiomatic language transformation.
- Exposed via asynchronous FastAPI endpoints for real-time, low-latency translation at scale supporting multiple languages.

University Assistant Chatbot (LangChain + OpenAI)

- Built a conversational AI assistant using LangChain and OpenAI GPT with persistent conversation memory, context-aware response generation, and a modular ingestion-retrieval-response pipeline.

Habit Arena — Gamified Habit Tracking Full-Stack App

- Built a full-stack app with FastAPI (SQLModel ORM), JavaScript frontend, JWT authentication, leaderboard engine, and gamification mechanics. Deployed on Render + Netlify with automated CI/CD.

Driver Drowsiness Detection (YOLOv8 + PyTorch)

- Trained a custom real-time object detection model with YOLOv8 and PyTorch to detect driver drowsiness from video frames, achieving over 90% detection accuracy at 30 FPS on standard hardware.

Certifications

- **Google Cloud Associate Cloud Engineer** | Google Cloud Platform (GCP) | Issued by Google
- **IBM Deep Learning Specialization** | IBM / Coursera | Neural Networks, PyTorch, Deep Learning Architectures
- **IBM AI Systems Design** | IBM / Coursera | AI Pipeline Design, Scalable AI Systems, Production Deployment

Education

Visvesvaraya National Institute of Technology (VNIT)

B.Tech in Mechanical Engineering | GPA: 8.2/10

Relevant coursework: Data Structures & Algorithms, Mathematics, Statistics, Computer Programming

Nagpur, India

Aug 2019 – May 2023